РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №11

дисциплина: Администрирование сетевых подсистем

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МОСКВА

2025 г.

Цель работы:

Целью данной работы является приобретение практических навыков по настройке удалённого доступа к серверу с помощью SSH.

Выполнение работы:

```
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server ~]# passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server ~]#
```

```
root@server:~
root@server:~    root@server:~
   Subject: A start job for unit systemd-hostnamed.service has begun execution
   Defined-By: systemd
   Support: https://wiki.rockylinux.org/rocky/support

   A start job for unit systemd-hostnamed.service has begun execution.

   The job identifier is 2680.
Mar 11 16:09:46 server.user.net systemd[1]: Started Hostname Service.
   Subject: A start job for unit systemd-hostnamed.service has finished successf
ully
   Defined-By: systemd
   Support: https://wiki.rockylinux.org/rocky/support

   A start job for unit systemd-hostnamed.service has finished successfully.

   The job identifier is 2680.
Mar 11 16:10:16 server.user.net systemd[1]: systemd-hostnamed.service: Deactivat
ed successfully.
   Subject: Unit succeeded
   Defined-By: systemd
   Support: https://wiki.rockylinux.org/rocky/support

   The unit systemd-hostnamed.service has successfully entered the 'dead' state.
```

Open ▾ | sshd_config /etc/ssh | Save

```
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to no to disable s/key passwords
```

Saving file "/etc/ssh/sshd_config"... | Plain Text ▾ | Tab Width: 8 ▾ | Ln 44, Col 1 | INS

```
[root@client ~]# ssh root@server.vagrant.net
ssh: Could not resolve hostname server.vagrant.net: Name or service not known
[root@client ~]# ssh root@server.root.net
ssh: Could not resolve hostname server.root.net: Name or service not known
[root@client ~]# ssh root@server@root.net
ssh: Could not resolve hostname root.net: Name or service not known
[root@client ~]# ssh root@server@vagrant.net
ssh: Could not resolve hostname vagrant.net: Name or service not known
[root@client ~]# ssh root@server
ssh: Could not resolve hostname server: Name or service not known
[root@client ~]# sudo -i
[root@client ~]# cd ..\exit
-bash: cd: ..exit: No such file or directory
[root@client ~]# exit
logout
[root@client ~]# ssh root@server
ssh: Could not resolve hostname server: Name or service not known
[root@client ~]# ssh root@server.root.net
ssh: Could not resolve hostname server.root.net: Name or service not known
[root@client ~]#
```

```
ssh: Could not resolve hostname server: Name or service not known
[root@client ~]# ssh root@server.root.net
ssh: Could not resolve hostname server.root.net: Name or service not known
[root@client ~]# ssh vagarnt@server.vagarnt.net
ssh: Could not resolve hostname server.vagarnt.net: Name or service not known
[root@client ~]# ssh vagrant@server.vagrant.net
ssh: Could not resolve hostname server.vagrant.net: Name or service not known
[root@client ~]#
```

Open *sshd_config /etc/ssh Save

```
# and KbdInteractiveAuthentication to 'no'.
# WARNING: 'UsePAM no' is not supported in RHEL and may cause several
# problems.
#UsePAM no

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
#X11Forwarding no
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
#PrintMotd yes
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /var/run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

        AllowUsers vagrant user

# override default of no subsystems
Subsystem       sftp    /usr/libexec/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#       X11Forwarding no
#       AllowTcpForwarding no
```

Plain Text Tab Width: 8 Ln 120, Col 13 INS

```
root@server:~
root@server:~                                  root@server:~
(gedit:6861): dconf-WARNING **: 16:57:30.437: failed to commit changes t
to execute child process “dbus-launch” (No such file or directory)
[root@server ~]# systemctl restart sshd
[root@server ~]# systemctl status -l sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; pres
     Active: active (running) since Wed 2026-03-11 16:57:35 UTC; 8s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 6902 (sshd)
      Tasks: 1 (limit: 22435)
     Memory: 1.4M
        CPU: 28ms
     CGroup: /system.slice/sshd.service
             └─6902 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 star
```

```
root@server:~
root@server:~                                  root@server:~
(gedit:6861): dconf-WARNING **: 16:57:30.437: failed to commit changes to dconf: Failed
to execute child process “dbus-launch” (No such file or directory)
[root@server ~]# systemctl restart sshd
[root@server ~]# systemctl status -l sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Wed 2026-03-11 16:57:35 UTC; 8s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 6902 (sshd)
      Tasks: 1 (limit: 22435)
     Memory: 1.4M
        CPU: 28ms
     CGroup: /system.slice/sshd.service
             └─6902 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 11 16:57:35 server.user.net systemd[1]: Starting OpenSSH server daemon...
Mar 11 16:57:35 server.user.net sshd[6902]: error: Bind to port 2022 on 0.0.0.0>
Mar 11 16:57:35 server.user.net sshd[6902]: error: Bind to port 2022 on :: fail>
Mar 11 16:57:35 server.user.net sshd[6902]: Server listening on 0.0.0.0 port 22.
Mar 11 16:57:35 server.user.net sshd[6902]: Server listening on :: port 22.
Mar 11 16:57:35 server.user.net systemd[1]: Started OpenSSH server daemon.
lines 1-18/18 (END)
```

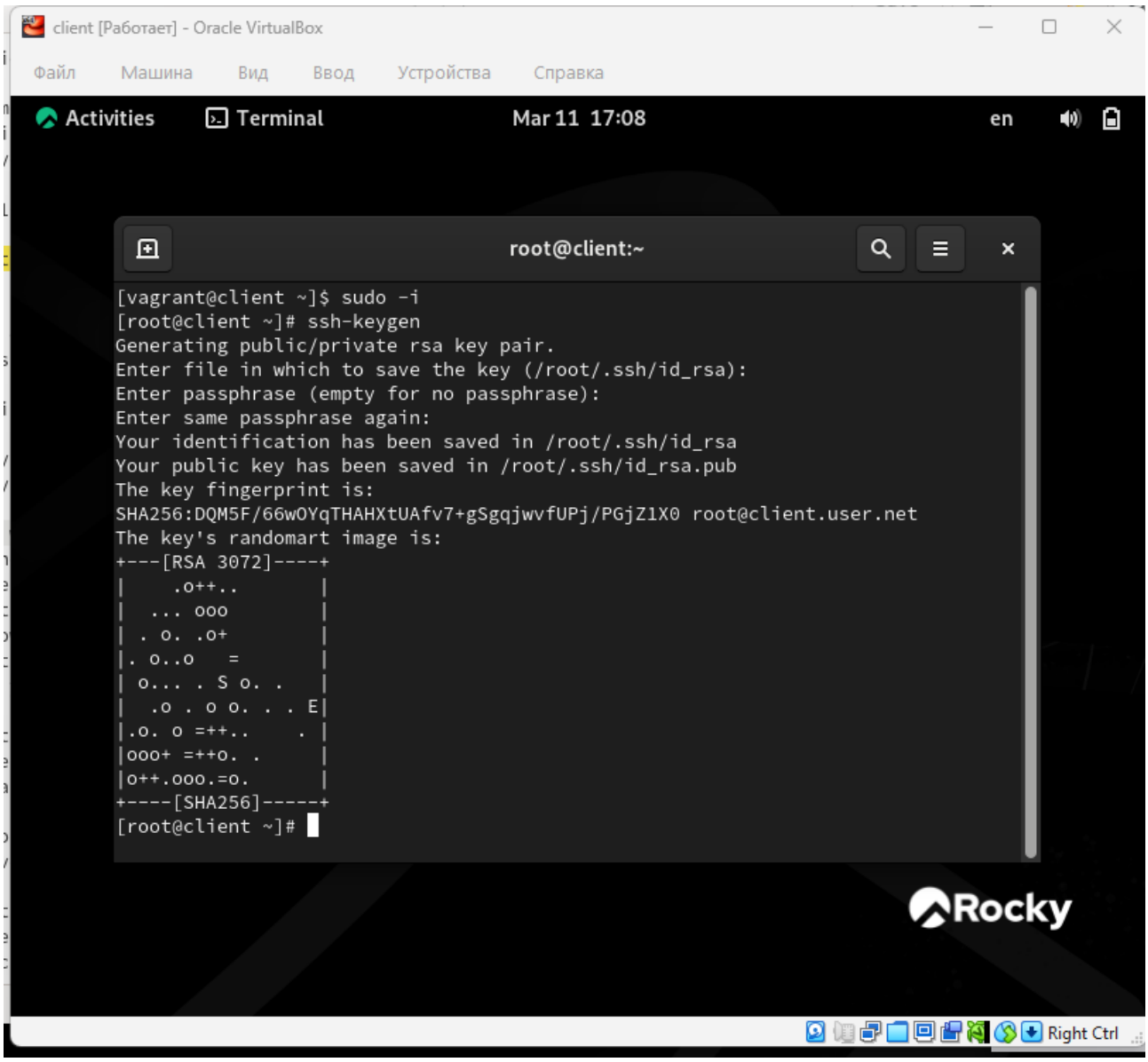
Open Save

sshd_config
/etc/ssh

PubkeyAuthentication 1 of 1

```
# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
```

Plain Text Tab Width: 8 Ln 46, Col 2 INS


client [Работает] - Oracle VirtualBox
Файл Машина Вид Ввод Устройства Справка
Activities Terminal Mar 11 17:08 en
root@client:~
[vagrant@client ~]$ sudo -i
[root@client ~]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:DQM5F/66wOYqTHAHXtUAfv7+gSgqjwvfUPj/PGjZ1X0 root@client.user.net
The key's randomart image is:
+---[RSA 3072]----+
| .o++.. |
| ... ooo |
| . o. .o+ |
|. o..o = |
| o... . S o. . |
| .o . o o. . . E|
|.o. o =++.. . |
|ooo+ =++o. . |
|o++.ooo.=o. |
+----[SHA256]-----+
[root@client ~]#
Rocky
Right Ctrl

```
vagrant@client:~
The key's randomart image is:
+---[RSA 3072]----+
|     .o++..       |
|  ... ooo         |
| . o. .o+         |
|. o..o   =        |
| o... . S o. .    |
|  .o . o o. . . E|
|.o. o =++..     . |
|ooo+ =++o. .      |
|o++.ooo.=o.       |
+----[SHA256]-----+
[root@client ~]# ssh-copy-id user@server.user.net
/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub
"
/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out
any that are already installed

/bin/ssh-copy-id: ERROR: ssh: Could not resolve hostname server.user.net: Name o
r service not known

[root@client ~]#
logout
[vagrant@client ~]$
```

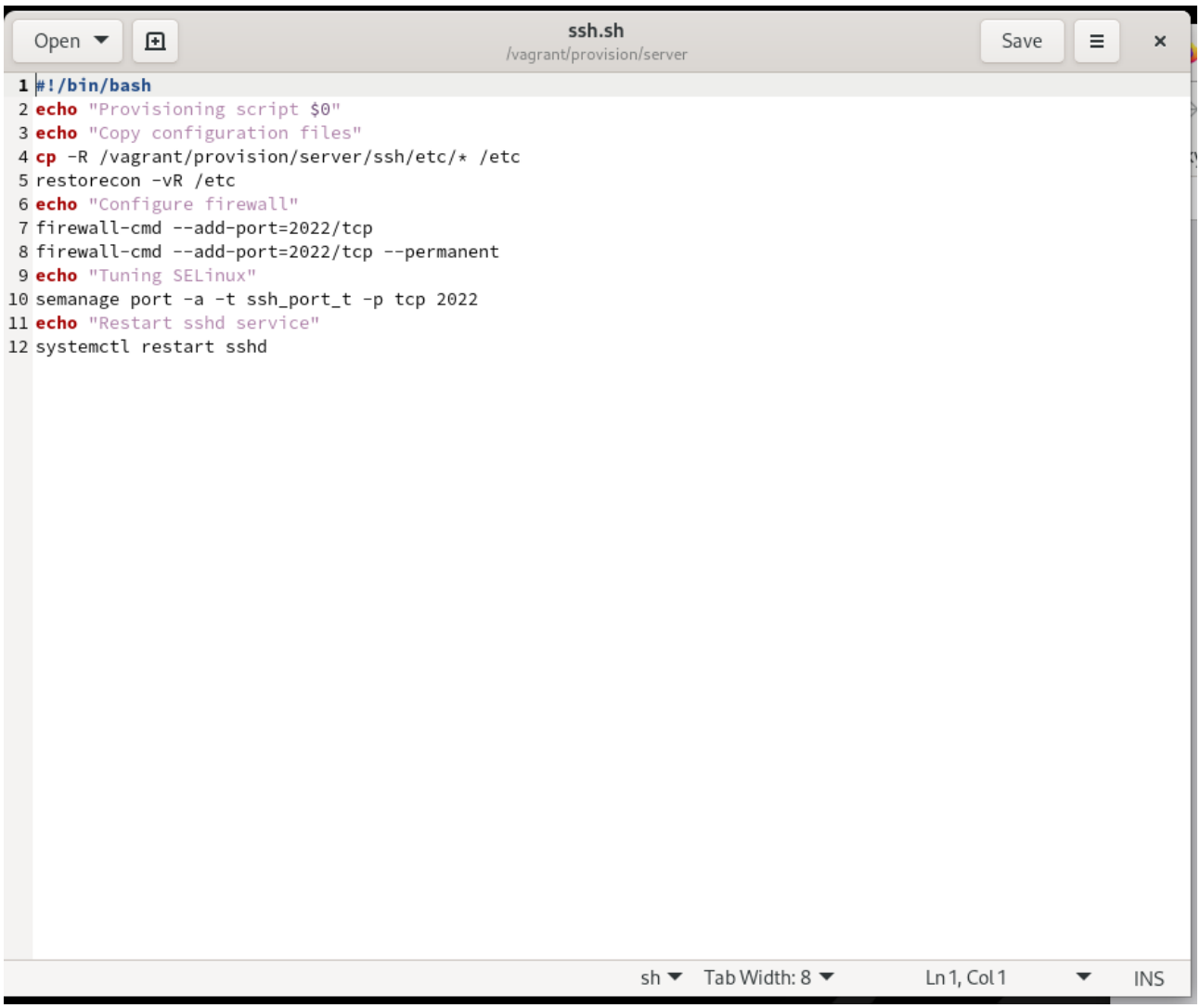
```
root@server:~
root@server:~    root@server:~    root@server:~
to execute child process “dbus-launch” (No such file or directory)

** (gedit:7031): WARNING **: 17:07:31.628: Set document metadata failed: Setting attribu
te metadata::gedit-spell-language not supported

** (gedit:7031): WARNING **: 17:07:31.630: Set document metadata failed: Setting attribu
te metadata::gedit-encoding not supported

** (gedit:7031): WARNING **: 17:11:37.098: Set document metadata failed: Setting attribu
te metadata::gedit-spell-language not supported

** (gedit:7031): WARNING **: 17:11:37.121: Set document metadata failed: Setting attribu
te metadata::gedit-encoding not supported

** (gedit:7031): WARNING **: 17:11:38.732: Set document metadata failed: Setting attribu
te metadata::gedit-position not supported

(gedit:7031): dconf-WARNING **: 17:11:38.761: failed to commit changes to dconf: Failed
to execute child process “dbus-launch” (No such file or directory)
[root@server ~]# ssh -fNL 8080:localhost:80 user@server.user.net
The authenticity of host 'server.user.net (fe80::a00:27ff:fe69:a8d%eth0)' can't be estab
lished.
ED25519 key fingerprint is SHA256:yiD6CwwpAVWgmZ0LNexnVt25Q77Gy5XU7NfNuqLLz68.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: server
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server.user.net' (ED25519) to the list of known hosts.
user@server.user.net's password:
```

Open ▾ *sshd_config /etc/ssh Save

```
# and KbdInteractiveAuthentication to 'no'.
# WARNING: 'UsePAM no' is not supported in RHEL and may cause several
# problems.
#UsePAM no

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
 X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
#PrintMotd yes
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /var/run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

        AllowUsers vagrant user

# override default of no subsystems
Subsystem       sftp    /usr/libexec/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
        X11Forwarding yes
```

Plain Text ▾ Tab Width: 8 ▾ Ln 130, Col 26 ▾ INS

```
user@server.user.net's password:
user@server.user.net: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,passwor
).
[root@server ~]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/ssh/etc/ssh
[root@server server]# cp -R /etc/ssh/sshd_config /vagrant/provision/server/ssh/etc/ssh/
[root@server server]# cd /vagrant/provision/server
[root@server server]# touch ssh.sh
[root@server server]# chmod +x ssh.sh
[root@server server]# gedit ssh.sh

(gedit:7114): dconf-WARNING **: 17:17:30.664: failed to commit changes to dconf: Failed
```



Ответы на контрольные вопросы:

1. Вы хотите запретить удалённый доступ по SSH на сервер пользователю root и разрешить доступ пользователю alice. Как это сделать? –

   В конфигурационном файле SSH /etc/ssh/sshd_config:

   # Запрет удалённого доступа пользователю root

   PermitRootLogin no

 # Разрешение доступа пользователю alice

AllowUsers alice

После внесения изменений, необходимо перезапустить службу SSH:

sudo service ssh restart

2. Как настроить удалённый доступ по SSH через несколько портов? Для чего это может потребоваться? **–**

   В конфигурационном файле /etc/ssh/sshd_config добавьте строки:

   # Первый порт (по умолчанию 22)

   Port 22

   # Второй порт

   Port 2222

   После изменений перезапустите службу SSH. Это может быть полезно для повышения безопасности, а также для избежания конфликтов с другими службами, использующими порт 22.

3. Какие параметры используются для создания туннеля SSH, когда команда ssh устанавливает фоновое соединение и не ожидает какой-либо конкретной команды? **-**

   ssh -N -f -L local_port:destination_host:remote_port user@ssh_server

   -N: Не выполнять команду на удаленном хосте.

   -f: Перевести ssh в фоновый режим после установки туннеля.

4. Как настроить локальную переадресацию с локального порта 5555 на порт 80 сервера server2.example.com? –

   ssh -L 5555:server2.example.com:80 user@ssh_server

   Теперь, при подключении к локальному порту 5555, трафик будет перенаправляться через SSH к порту 80 на сервере server2.example.com.

5. Как настроить SELinux, чтобы позволить SSH связываться с портом 2022? –

   sudo semanage port -a -t ssh_port_t -p tcp 2022

   Данная команда добавляет правило SELinux, разрешая использование порта 2022 для сервиса ssh.

6. Как настроить межсетевой экран на сервере, чтобы разрешить входящие подключения по SSH через порт 2022? –

   sudo firewall-cmd --permanent --add-port=2022/tcp

   sudo firewall-cmd --reload

   Эти команды добавляют правило в межсетевой экран для разрешения входящих подключений по SSH через порт 2022 и перезагружают конфигурацию межсетевого экрана.